Harish Kumar S

Associate Software Developer | AI/ML · MLOps · Data Platforms

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SUMMARY

Associate Software Developer at FactEntry (A SIX Company) building production AI and data platforms. Experienced in FastAPI microservices, Redis-based distributed processing, Kubernetes (K3s) with GitOps delivery, RAG systems, and computer vision, and now cloud data engineering on Microsoft Azure and Databricks (PySpark, data lake, data warehouse, MLflow). Integrated M.Tech in Data Science from VIT (2025).

EXPERIENCE

Associate Software Developer
Dec 2025 – Present

FactEntry Data Solutions (A SIX Company)

- Build and run the Data Mining Engine (DME), a production bond-extraction platform for corporate and municipal bonds on K3s (80+ CPU cores) with a human-in-the-loop validation UI.
- Designed Redis Streams job dispatch across multi-node workers, plus FastAPI microservices with live WebSocket job telemetry, role-based CPU tiering, and domain-restricted auth.
- Implemented GitOps delivery: GitLab CI image builds, Helm chart promotion, and Argo CD sync to staging.
- Working with SIX (parent company) to migrate the DME data-mining architecture onto Microsoft Azure and Databricks: PySpark, data lakes/warehouses, Azure data engineering, and MLflow for experiment and model tracking. Received the SIX Excellence Award for this work.

Associate Software Engineer Trainee Intern
Jun 2025 – Oct 2025

MIMASOFT Technologies

- Fine-tuned two YOLOv11 detection models in Roboflow for dental imaging: tooth detection (mAP@50 87.7%, precision 89.3%) and 14-class panoramic X-ray (mAP@50 70.3%).
- Built an inference CLI for Roboflow serverless (inference-sdk) and local model backends; contributed to backend API integration.

MLOps and Data Science Intern
Nov 2024 – May 2025

FactEntry Data Solutions (A SIX Company)

- Published language-processing-tool on PyPI (v0.2.7) for PDF language detection, OCR, and layout-based document clustering; delivered an advanced internal version (VGG16/ResNet clustering, LLaMA Vision) to FactEntry.
- Built Fixed Income Expert, a RAG chatbot over bond documents (SBERT, FAISS, Groq LLaMA 3, FastAPI + Streamlit), deployed as systemd services; contributed to DME prototype R&D.

Data Science Intern (ML and SDE)
Nov 2023 – Jan 2024

JP INFOTECH

- Built an extractive summarization and information-retrieval system using NLP-driven web scraping with English-to-SQL query translation (SQLite/MySQL), reaching 95% extraction accuracy.
- Built an applicant tracking tool that OCRs resumes (PyTesseract, OpenCV), matches them to job descriptions, and suggests missing keywords.

Applied Data Science Extern
May 2023 – Jul 2023

Smart Bridge - SmartInternz (Google Developers)

- Built a Random Forest + SVM ensemble for disease prediction on clinical-trial data, fed by a MongoDB Atlas (PyMongo) data pipeline.

PROJECTS

Data Mining Engine (DME) | K3s, FastAPI, Redis, PostgreSQL, Argo CD, Azure, Databricks, PySpark, MLflow
Enterprise bond extraction on K3s with GitOps CI/CD; migrating to Microsoft Azure and Databricks with SIX (PySpark, data lake, warehouse, MLflow). SIX Excellence Award.

Recruiting Agent — AI-Powered ATS | FastAPI, React, SQLite, ChromaDB, Ollama, Groq
Nine-step LLM ATS: resume OCR, ChromaDB JD matching, claim verification, fit and authenticity scoring, HR questions.

Trip Guide — Conversational Trip Planner | FastAPI, Google ADK, Gemini, Groq, Ollama, Leaflet
Google ADK multi-agent trip planner (weather, places, stays agents) with SSE chat, live trip board, and OSRM map.

Cybersecurity Content Moderator — VIT Capstone | Ollama, FAISS, FastAPI, Streamlit, Docker, GitHub Actions
Multimodal moderation of text, images, and PDFs with custom Ollama models (WizardLM 7B, Gemma 3 12B vision), FAISS retrieval over prior cases to ground verdicts, GPU inference, and CI-published Docker images.

PAI PDF AI — Document Intelligence Platform | FastAPI, React, Elasticsearch, Ollama, PyMuPDF
Privacy-first PDF analysis with local LLMs: clause, entity, and risk extraction with answers cited to source chunks.

SKILLS

Languages & Backend: Python, SQL, FastAPI, Flask, WebSockets, React, Streamlit

AI / ML & GenAI: PyTorch, TensorFlow, scikit-learn, Deep Learning, NLP, Computer Vision, RAG, Agentic AI, n8n, LLM Agents (Google ADK), LangChain, FAISS, ChromaDB, Sentence Transformers, Ollama (fine-tuning), Groq, YOLOv11, Roboflow, OpenCV, Tesseract OCR

Data & Cloud: Databricks, PySpark, Delta Lake, Azure Data Engineering, Data Lake, Data Warehouse, ETL, Microsoft Azure, PostgreSQL, Redis Streams, MongoDB, Elasticsearch

MLOps & DevOps: Kubernetes (K3s), Docker, Helm, Argo CD, GitLab CI, GitHub Actions, GitOps, MLflow, Linux, GPU Computing

EDUCATION

Integrated M.Tech, Computer Science and Engineering (Data Science) 2020 – 2025

Vellore Institute of Technology | CGPA 7.8

Class XII: 79%, THIRU G V C Higher Secondary School (2020) | Class X: 87.8%, St Joseph Matriculation Higher Secondary School (2018)

CERTIFICATIONS

Awards and exam credentials

- SIX Excellence Award — DME Azure/Databricks Migration — SIX, Oct 2026
- Microsoft Certified: Azure AI Fundamentals — Microsoft, Jun 2023
- Applied Data Science (SmartInternz + Google Developers) — SmartInternz / Google Developers, 2023

Course certificates

- Complete Guide to Databricks for Data Engineering — LinkedIn Learning, Oct 2026
- Microsoft Azure Essentials Professional Certificate — Microsoft / LinkedIn Learning, Aug 2026
- Build Multi-Agent Systems with n8n — Orchestrating Reliable AI Workflows — LinkedIn Learning, 2026
- Data Analysis and Visualization with Power BI — Microsoft / Coursera, 2025
- Preparing Data for Analysis with Microsoft Excel — Microsoft / Coursera, 2024
- Harnessing the Power of Data with Power BI — Microsoft / Coursera, 2024
- Extract, Transform, and Load Data in Power BI — Microsoft / Coursera, 2024